# Assignment :- 2

Name: Kaushal kumar Ray
PRN: 240205241023
Branch: B.Tech CSE with AI & ML
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① Analyze different project Estimation Techniques.

→ Project estimation helps predict time, cost, and resources needed
Common Techniques :-

a) Expert judgment
- Based on experience of senior developers / managers
- Fast but may be biased.

b) Analogous Estimation
- Uses data from similar past projects
- Quick but less accurate

c) Parametric models
- Uses mathematical formulas
- Based on size

d) Bottom-up Estimation

- Break project into tasks → estimate each → sum up.
- Highly accurate but time-consuming

e) Agile Estimation

- uses relative sizing
- flexible and iterative.

② Explain Risk management process with a diagram.

→ Risk management identifies and control potential problems.

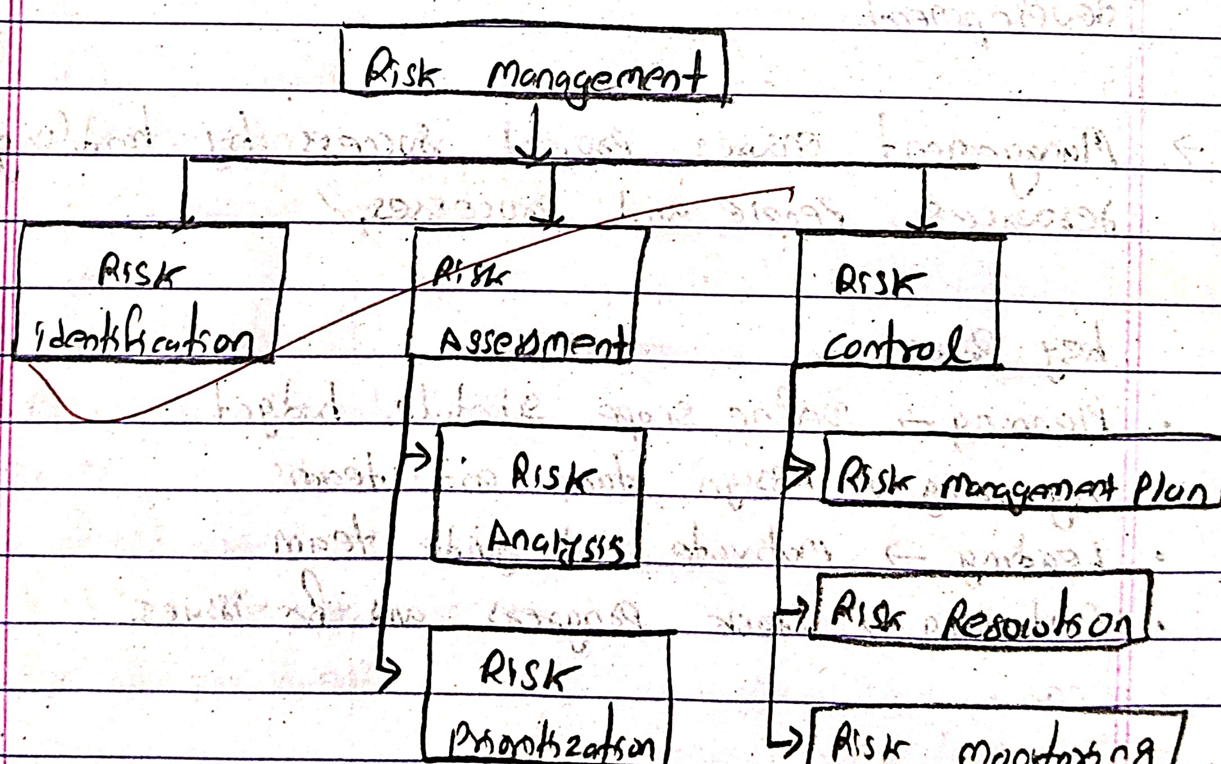



fig: Risk Management Process

Steps:

1) Risk identification
11) Risk Analysis
111) Risk prioritization
1v) Risk mitigation planning
v) Risk monitoring

Example:

- Risk: Developer leaves project
- Mitigation: knowledge sharing, documentation.

③ Analyze the role of management in software development.

→ Management ensures project success by handling resources, people and processes.

key Roles:-

- Planning → Define scope, schedule, budget
- Organizing → Assign tasks and teams
- Leading → motivate and guide team
- Controlling → Track progress and fix issues.

Responsibilities:
- Risk handling
- Quality assurance
- communication with stakeholders
- Decision making

(4) Explain Software Process Metrics and project metrics.

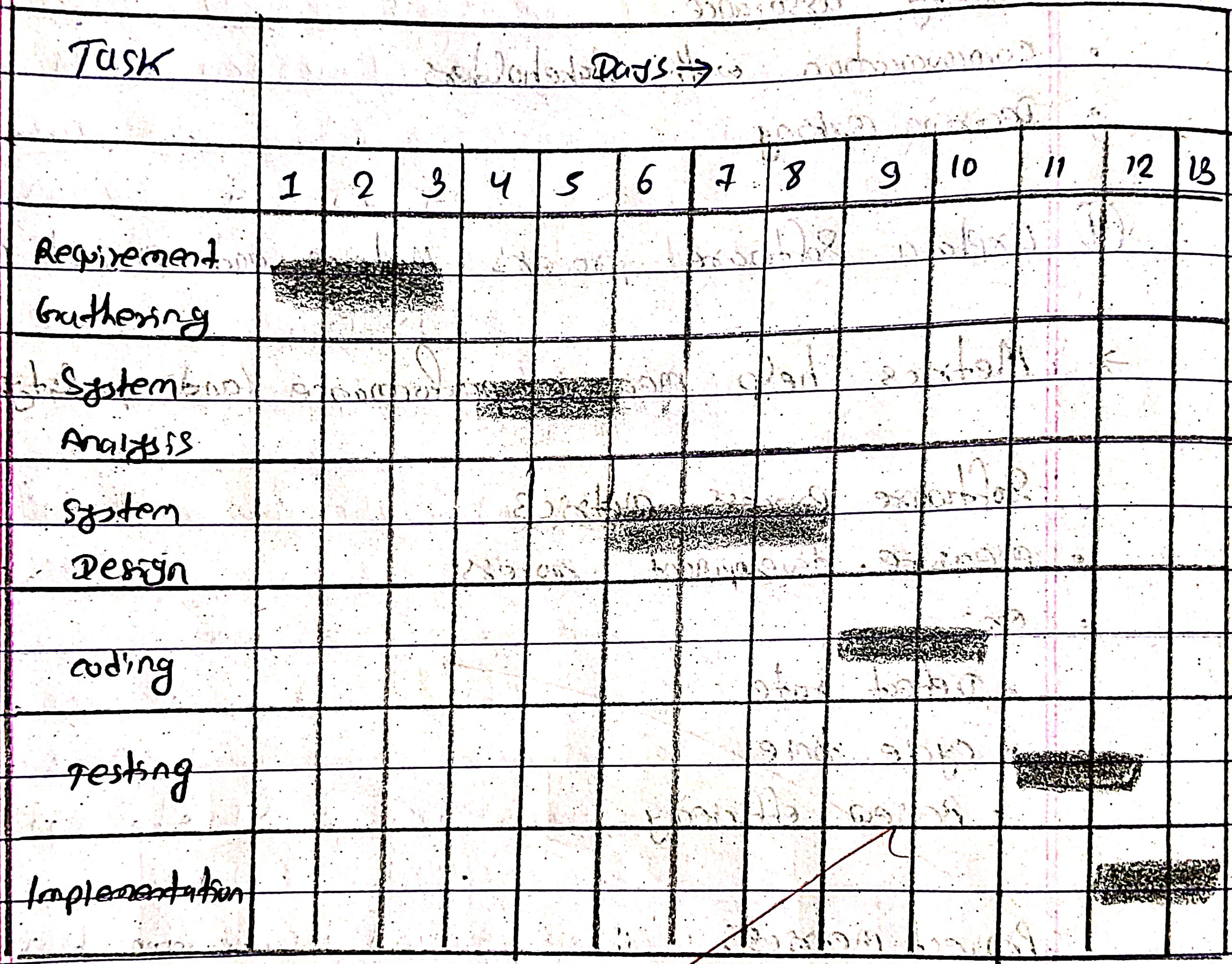
→ Metrics help measure performance and quality.

Software Process metrics
- measure development process
- ex:
  - Defect rate
  - cycle time
  - Review efficiency

Project metrics:
- Measures project status
  - ex:
  - cost variance
  - schedule variance
  - Effort spent

Difference
- Process metrics → improve workflow
- Project metrics → track progress

⑤ Explain Software project scheduling using Gantt chart.




Features:

- Horizontal bars represent task
- Timeline on x-axis
- Shows dependencies and duration

Benefits:

- Easy visualization
- Helps track progress
- Identifies delays

(6) Discuss Earned Value Analysis with a numerical example

→ Earned Value Analysis (EVA) is a technique used to measure project performance by comparing:

- Planned work – Planned value (PV)
- Actual work completed – Earned value (EV)
- Actual cost – Actual cost (AC)

It helps determine whether a project is on schedule and within budget.

Formulas:-

i) Cost variance (CV)

$CV = EV - AC$

- $CV > 0$ → under budget
- $CV < 0$ → over budget

ii) Schedule variance (SV)

$SV = EV - PV$

- $SV > 0$ → ahead of schedule
- $SV < 0$ → Behind schedule

iii) Cost performance index (CPI)

$$CPI = \frac{EV}{AC}$$

- $CPI > 1$ → cost efficient
- $CPI < 1$ → cost overrun

iv) Schedule performance Index (SPI)

$$SPI = \frac{EV}{PV}$$

- $SPI > 1$ → ahead of schedule
- $SPI < 1$ → behind schedule

Example :

- Planned value (PV) = 100000
- Earned value (EV) = 80,000
- Actual cost (AC) = 90,000

Sow,

i) Cost variance = EV - AC
= 80,000 - 90000
= -10,000

Project over budget

ii) Schedule variance (SV) = EV - PV
= 80000 - 100000
= -20000

Project is behind schedule

iii) Calculate CPI = 80000/90000
= 0.89

cost efficiency is low (overspending)

iv) calculate SPI = 80,000 / 100000 = 0.8

progress is slower than planned

- over budget (CV < 0, CPI < 1)
- Behind schedule (SV < 0, SPI < 1)

⑦ Discuss the SCM process and tools.

→ Software configuration management manages changes in software.

SCM Process:

i) Configuration Identification
ii) Version control
iii) Change control
iv) Status Accounting
v) Configuration Audit

Tools:

- Git → version control
- SVN → centralized version control
- Jenkins → CI/CD integration
- Docker → Environment consistency

SCM ensures:

- No data loss
- Proper version tracking
- Controlled changes.